Scrum

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Scrum



Scrum

- Scrum is a position in rugby in which players from two competing teams pack together closely to gain control of the rugby ball.
- In project management, however, scrum is a lightweight, iterative, and incremental agile process model.
- That means, scrum prioritizes simplicity in a project and develops product in a gradual manner with frequent delivery.
- Currently, scrum is one of the widely used process models in software industry.
- The word **“Scrum”** originally comes from **rugby**, where it refers to a **formation of players** who pack closely together with their heads down and attempt to gain possession of the ball.
- It symbolizes **teamwork**, **coordination**, **communication**, and **collective effort** — everyone pushes together toward a common goal.

Pillars of Scrum

Transparency

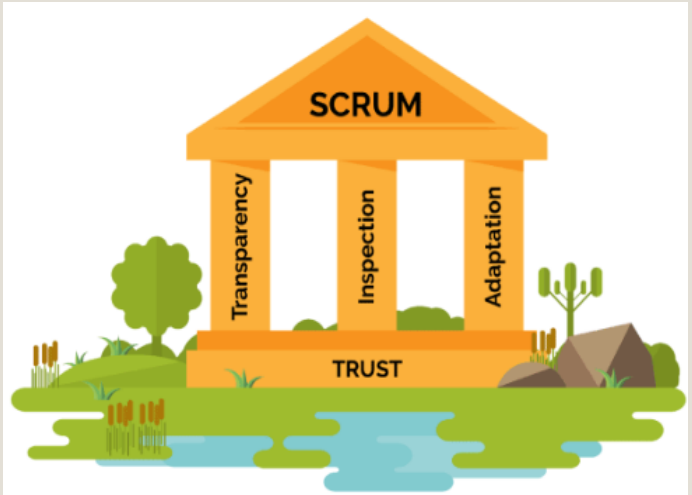
- All processes and progress must be visible to everyone involved.
- Artifacts like **backlogs** help maintain openness.

Inspection

- Regularly reviewing progress helps identify problems or deviations early.
- Inspections occur during meetings like **Daily Scrum** and **Sprint Review**.

Adaptation

- Based on inspection results, the team adjusts processes or goals to improve outcomes.
- Adaptation occurs in **Sprint Retrospectives** or when backlog priorities change.



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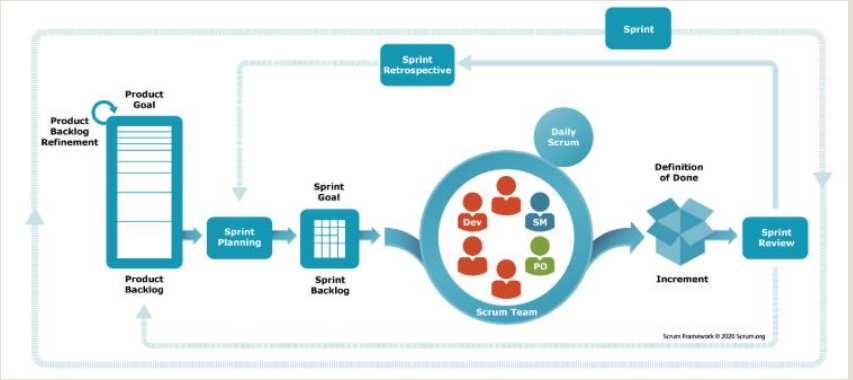
Sprint and Increment

Sprint

- The core timeboxed period (usually **1 month**) where a usable product increment is created.
- Has a fixed duration and consistent schedule.
- Each Sprint includes all other Scrum events.

Increment

- The **working product output** produced at the end of a Sprint.
- Must meet the **Definition of Done (DoD)** and be potentially shippable.



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Scrum Master

- Is responsible and accountable for the proper execution of **Scrum**
- Ensures the Scrum framework is properly **understood** and **followed**.
- Removes obstacles that hinder the team's progress.
- Acts as a facilitator, not a boss — supports collaboration and efficiency.
- Scrum master is usually the **project manager.**



Scrum Master

Helps the Scrum Team:

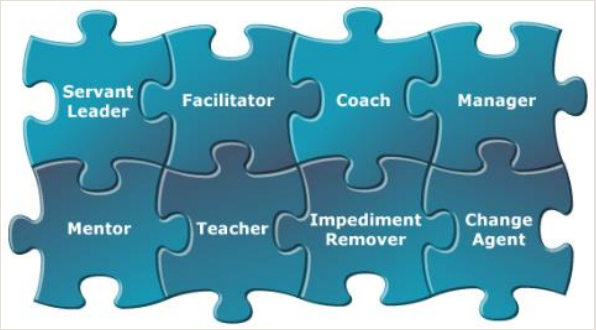
- By coaching the team members in self-management and cross-functionality
- Focus on creating high-value Increments that meet the Definition of Done
- Influence the removal of impediments to the Scrum Team's progress
- Ensure that all Scrum events take place and are positive, productive, and kept within the time-box.

Helps the Product Owner:

- Find techniques for effective Product Goal definition and Product Backlog management
- Provide ways for the Scrum Team to understand the need for clear and concise Product Backlog items
- Facilitate stakeholder collaboration as requested or needed

Supports the Organization:

- By Leading, training and coaching them in their Scrum adoption
- Remove barriers between stakeholders and Scrum Teams



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Product Owner

- Is responsible and accountable for the **quality of the product**.
- Represents the **customer or stakeholder** interests.
- Manages and prioritizes the **Product Backlog**.
- Defines clear goals and **ensures the team delivers maximum product value**.



Main Task:

- Developing and explicitly communicating the Product Goal
- Creating and clearly communicating Product Backlog Items
- Ordering Product Backlog Items
- Ensuring that the Product Backlog is transparent, visible and understood

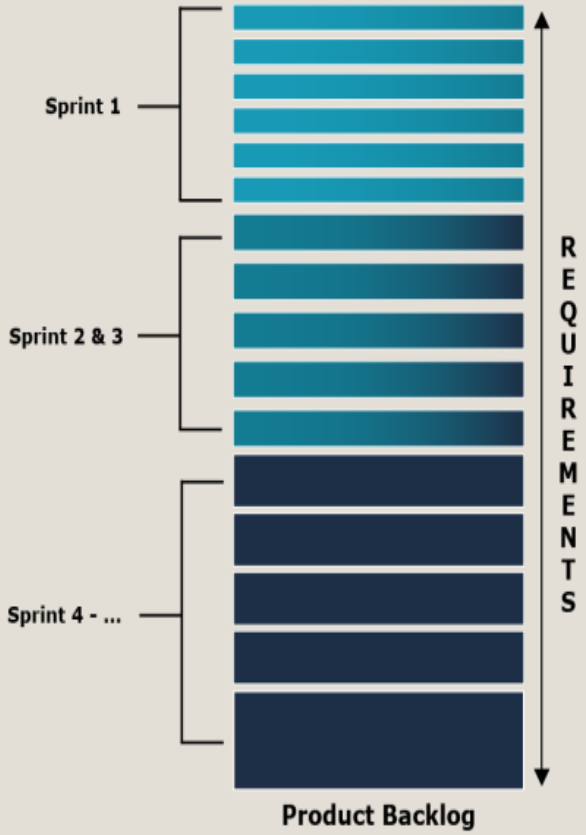
Product Backlog and Product Goal

Product Goal

- The **long-term objective** for the product.
- Provides direction for all Sprints and aligns the team's work.

Product Backlog

- An ordered list of **all desired features, fixes, and tasks** for the product.
- Continuously refined by the Product Owner.
- Commitment : **Product Goal**
- Product Backlog **refinement** is the act of breaking down and further defining Product Backlog items into smaller more precise items.



Scrum Overview

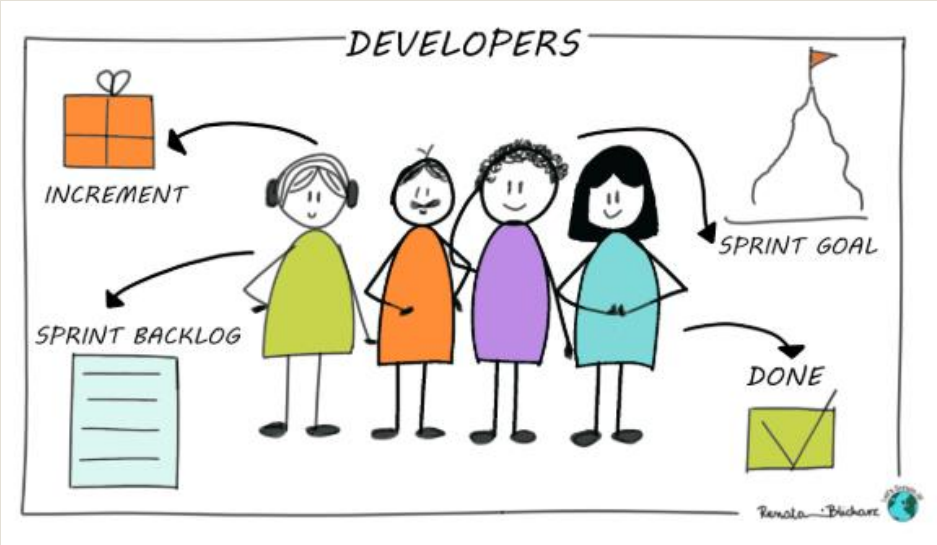
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Developers

- The professionals who **design, code, test, and deliver** the product increment.
- Responsible for achieving the **Sprint Goal**, the target of the Sprint.
- Create plan for the Sprint, the **Sprint Backlog**, and build an increment according to the plan
- Other roles: Facilitation, mentoring, teaching, coaching



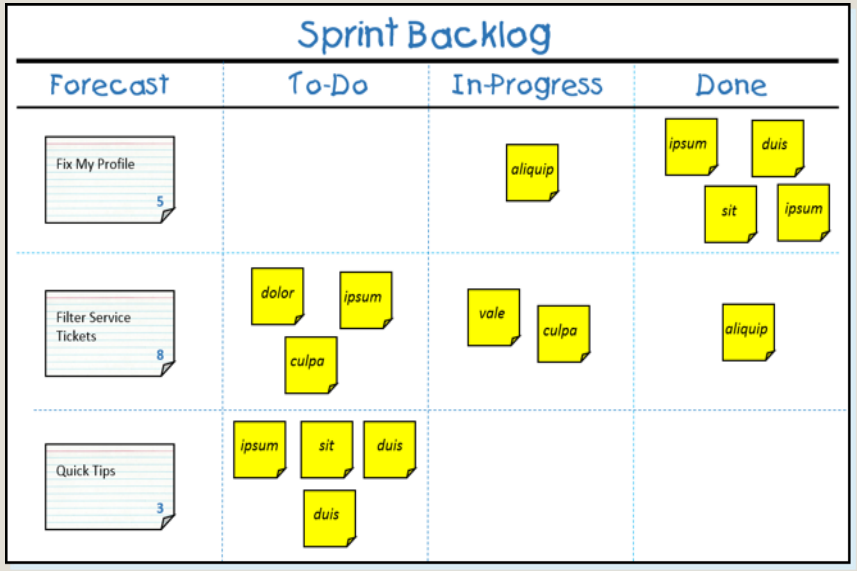
Sprint Backlog and Sprint Goal

Sprint Goal

- A **short-term objective or target** for the Sprint that guides the development team's focus.
- Set by the developers during **Sprint Planning**

Sprint Backlog

- A **subset of the Product Backlog** chosen for the current Sprint.
- Commitment : **Sprint Goal**
- Includes all tasks needed to achieve the Sprint Goal.



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Sprint Planning

- Held at the beginning of a Sprint.
- Defines **what** will be done (**Sprint Goal**) and **how** it will be achieved.
- The **Product Owner** presents the top-priority items, and the **Developers** selects tasks.
- The **Sprint Backlog** is developed from the selected **Product Backlog** items.
- Timeboxed to a maximum of **8 hours**.



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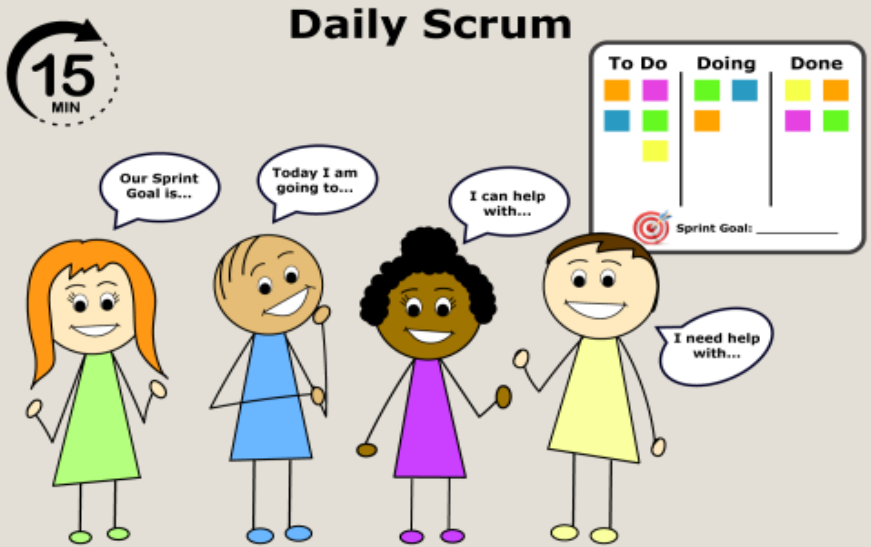
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Daily Scrum

- A short (**15-minute**) daily meeting for Developers.
- Team members discuss progress, upcoming work, and blockers.
- Sprint backlog items are discussed, and refined if necessary.
- Promotes coordination and keeps everyone aligned.



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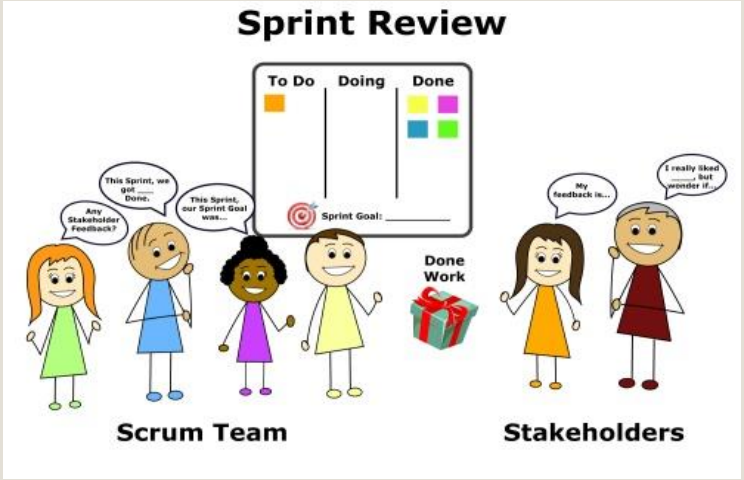
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Sprint Review



Sprint Review

- Held at the end of the Sprint to **inspect the increment**.
- The team demonstrates the new features to stakeholders.
- Stakeholders and Scrum team members are invited by the Product Owner
- Feedback is collected to improve the product backlog.
- Discusses which **Product Backlog** items are **Done** and which are **Not Done**.
- Discusses what problems the team faced, how they were resolved or not
- Review of product budget, timeline, marketplace demand, potential capabilities
- Timeboxed to a maximum of **4 hours**.



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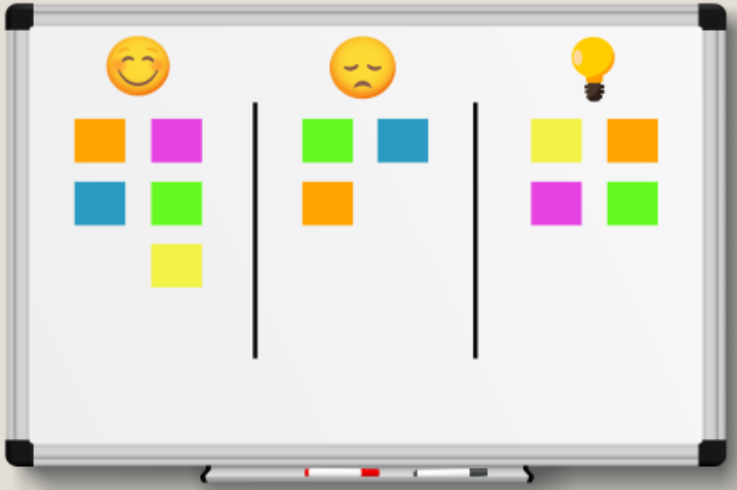
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Sprint Retrospective

- Final event of a Sprint.
- A short meeting of the Scrum Team after demonstration to the stakeholders.
- The team reflects on what went well, what didn’t, and how to improve next Sprint.
- Timeboxed to a maximum of **3 hours**.
- Focuses on continuous improvement in teamwork and processes.
- Gives input for the next Sprint Planning.
- **Sprint Review** – Improves Product
  Sprint Retrospective – Improves Process, Teamwork



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Definition of Done

- A clear **checklist or standard** that defines when a product increment is considered complete.
- Ensures **quality** and **consistency** across Sprints.
- Commitment for the **Increment**.
- **Example:** DoD for a health-focused software application

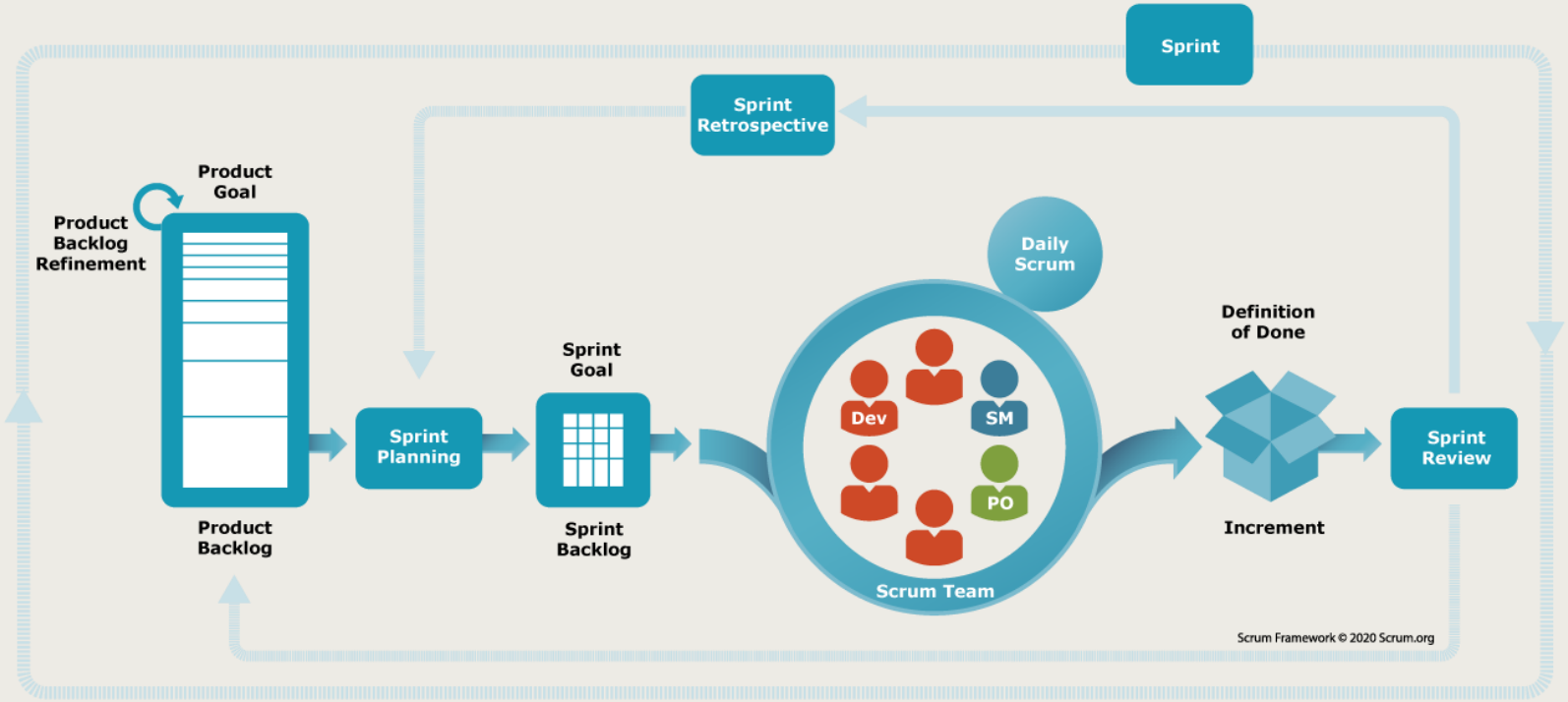
    1. All testing completed
    2. No known defects
    3. Code review completed and passed
    4. Meets HIPAA Compliance standards
    5. Meets general security requirements



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ICE BOX
EMERGENCY
IN PROGRESS
TESTING
COMPLETE
And that, gentlemen, is SCRUM.

References

- Slides adapted from Md. Muhaiminul Islam Nafi, Lecturer, Dept. of CSE, UIU
- What is Scrum? | Scrum.org


Thank
You